As mentioned, Drools likes to work with strongly typed values, so you can wrap each collection of attributes in a distinct class to make it simpler to write rules that match each one, as shown in listing 8.12. Open DroolsAccessController.java in your editor again and add the four wrapper classes from the following listing as inner classes to the `DroolsAccessController` class.

Listing 8.12 Wrapping attributes in types

```
public static class Subject extends HashMap<String, Object>
{    #A
    Subject(Map<String, Object> m) { super(m); }    #A
}    #A

public static class Resource extends HashMap<String, Object>
{    #B
    Resource(Map<String, Object> m) { super(m); }    #B
}    #B

public static class Action extends HashMap<String, Object> {
    Action(Map<String, Object> m) { super(m); }
}

public static class Environment extends HashMap<String,
Object> {
    Environment(Map<String, Object> m) { super(m); }
}
```

#A Wrapper for subject-related attributes.
#B Wrapper for resource-related attributes.

You can now start writing access control rules. Rather than re-implementing all the existing RBAC access control